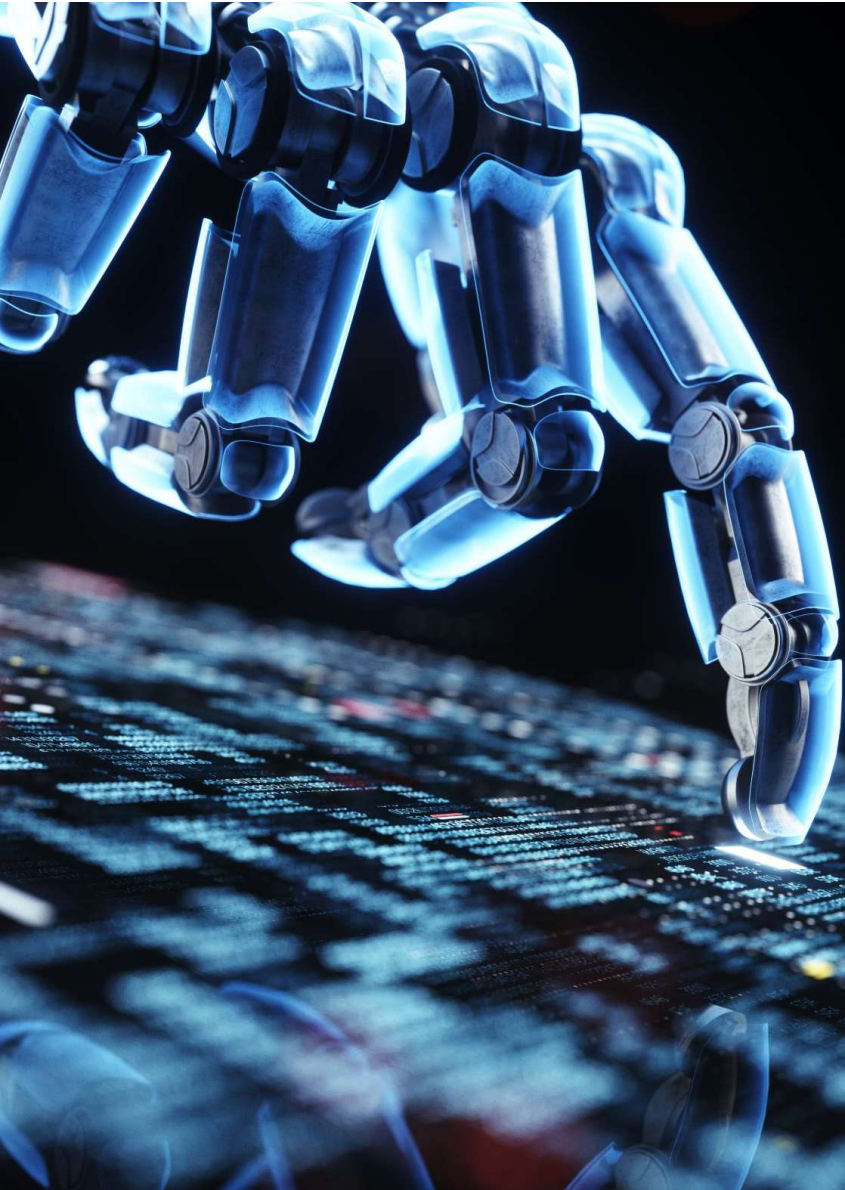




ProComp: A Problem-Driven Multi-Agent Framework for Comparative Paper Ranking

MASTER'S THESIS BY LAXMI SHRAVANI MAMIDALA

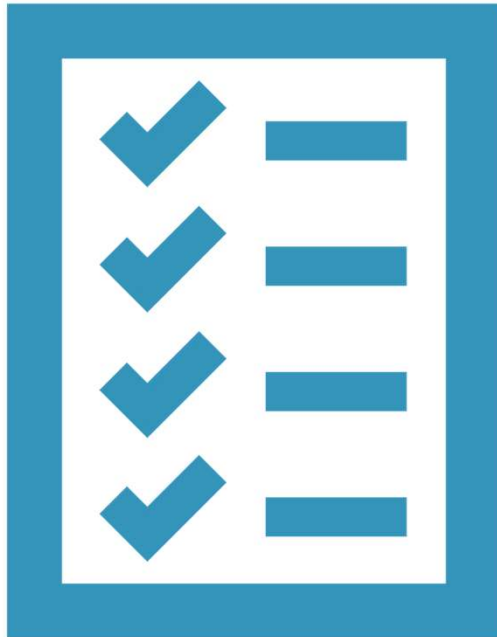
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AGENDA



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Background

► Evolution of Peer Review

The concept of peer review began in the 17th century as an informal mechanism for assessing academic research findings. It has developed to a complex regulatory system involving more than one layer of responsibility. With this framework in mind, scientific peers have a means of checking whether research at the academic level meets the highest standards.

► Bias and Subjectivity

Despite anonymity, the reviewer's evaluation is affected by subjective factors such as language, perceived innovation, or the institute's status. Many of these factors influence the review process and can lead to bias.

► Scalability Challenges

As scientific publishing has exploded, the peer review system hasn't kept up. Most reviewers are simply burned out and falling behind on their deadlines for too long. As peer review is often an add-on from the authors, with no extra time for them to get proper attention to their research, and no time for them to check it out as well.

► Inconsistency in Evaluations

Such low inter-reviewer reliability suggests that the validity and reliability of the evaluation criteria may be subjective rather than objective. This is undermining the independence and authority of a peer review system.

Limitations of Peer Review Systems

Reviewer Fatigue

The exponential increase in manuscript submissions has resulted in many reviewers facing mental and time constraints in their review process, leading to much longer evaluation times, much less investigation time, and therefore a lower-quality peer-review process.

Inconsistency in Evaluation

Variability in reviewer skill levels and subjective interpretation in the analysis can lead to different review scores and feedback for similar papers, resulting in inconsistencies and limited reliability across peer-review scores

Subjectivity and Bias

Writing style, perceived novelty, and institutional affiliation are two aspects of reviewers that can bias them (resulting in biased decisions) in the peer-review process.

Research problem

- ▶ The current peer-review system is constrained by limited scalability, inherent subjectivity, and a lack of standardized evaluation protocols, leading to inconsistent, biased, and non-reproducible assessment outcomes, thereby impeding the fairness, reliability, and efficiency of scientific manuscript evaluation.

Role of AI in Scientific Evaluation



Introduction to LLMs

Large language models (LLMs) like GPT and LLaMA are deep neural models, trained on high-level scientific corpora so that they perform semantic comprehension, contextual summarization and reasoning over challenging research.



Applications in Scientific Analysis

Large Language Models (LLMs) empower automated literature analysis: the study of literature can not only extract key insights but discover its methodological limitations and construct structured evaluations and thereby reduce reviewers' workload and enhance the quality of their assessment to better demonstrate the reliability of its insights as well.



Enhancing Peer Review

With the help of such experts as expert reviewers, LLMs are able to generate rapid, standardized feedback with both quantitative and qualitative evaluations, which facilitates the way data is processed and peer-review is robust.

Challenges in Traditional Peer Review Systems

Lack of Structural Decomposition

In single-agent models, all aspects of a manuscript are evaluated under one's own evaluation simultaneously, and shallow and general assessments are made. Without the need for task decomposition to form special parts, analysis and understanding of a manuscript are much less simple, interpretable, and thorough.

Hallucination and Limited Context

Such models may produce plausible but factually incorrect outputs due to poor domain grounding and inadequate integration of external knowledge, resulting in low assessment accuracy.

Poor Ranking and Discrimination

However, in the absence of structured scoring mechanisms and a joint evaluation system or other mechanisms (from peer-review methods), single-agent systems lack the capability to objectively evaluate research findings, leading to biased ranking and limited effectiveness for peer-review purposes.

Motivation for PROCOMP Framework

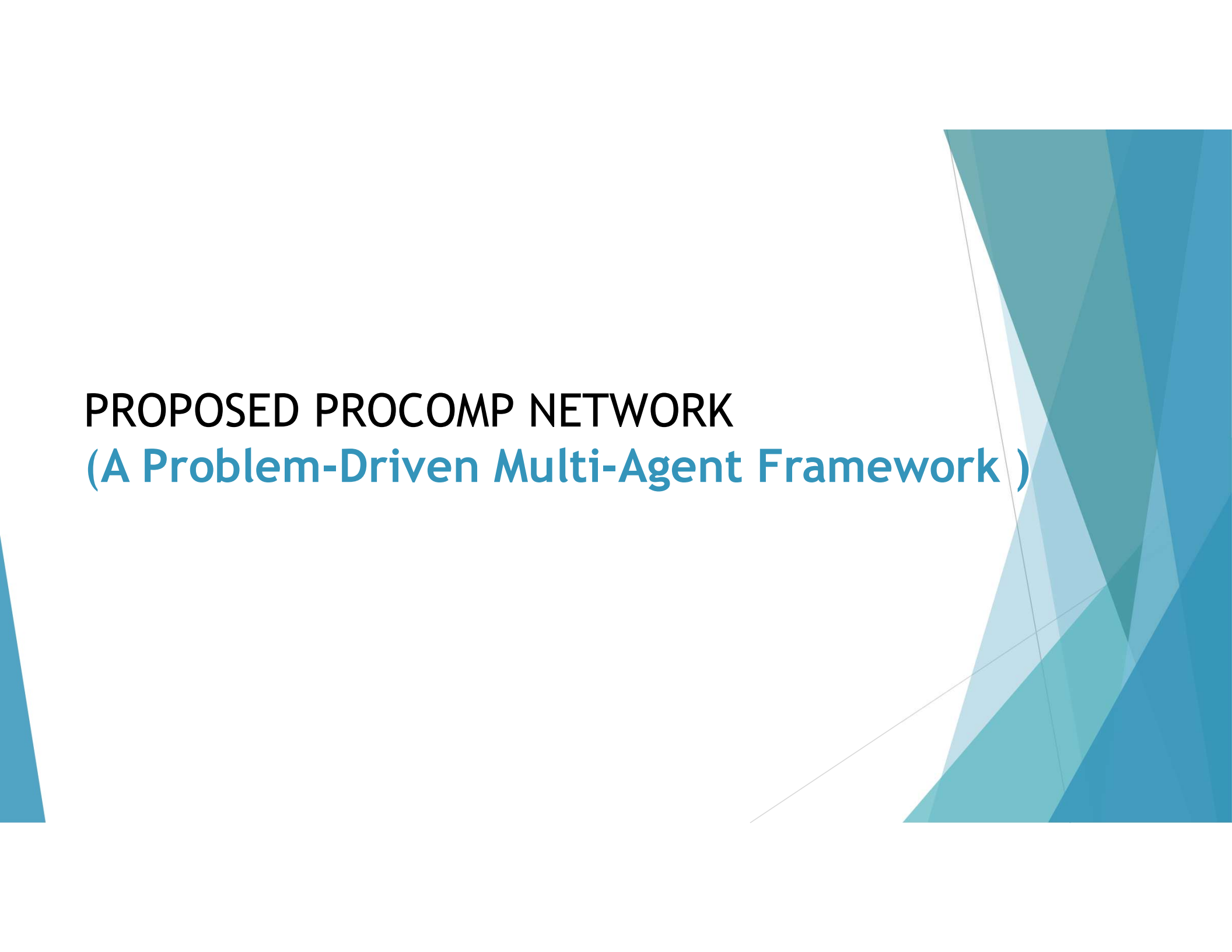


- ▶ The growth in the number of scientific publications has pushed peer review systems to the brink, with reviewers feeling fatigued and unreliable.
- ▶ We established ProComp because we need a system that serves as a framework for high-volume, science-written reviews with ethical, robust evaluation. ProComp uses expert professionals to test diverse aspects of the research, thereby minimizing bias and variation.
- ▶ ProComp provides a structured results system that lets readers compare relevant information and make quick decisions, making science processes more manageable and reliable. In practice, human reviewers have time to make accurate decisions.

Objectives

- ▶ Multi-Agent Decomposition
- ▶ Structured Review Outputs
- ▶ Retrieval-Augmented Reasoning
- ▶ Consistency and Robustness
- ▶ Scalability and Flexibility





PROPOSED PROCOMP NETWORK (A Problem-Driven Multi-Agent Framework)

Overview of ProComp Framework



Modular Multi-Agent Pipeline



Retrieval and Meta-Review Integration

Multi-Agent ProCOMP Architecture

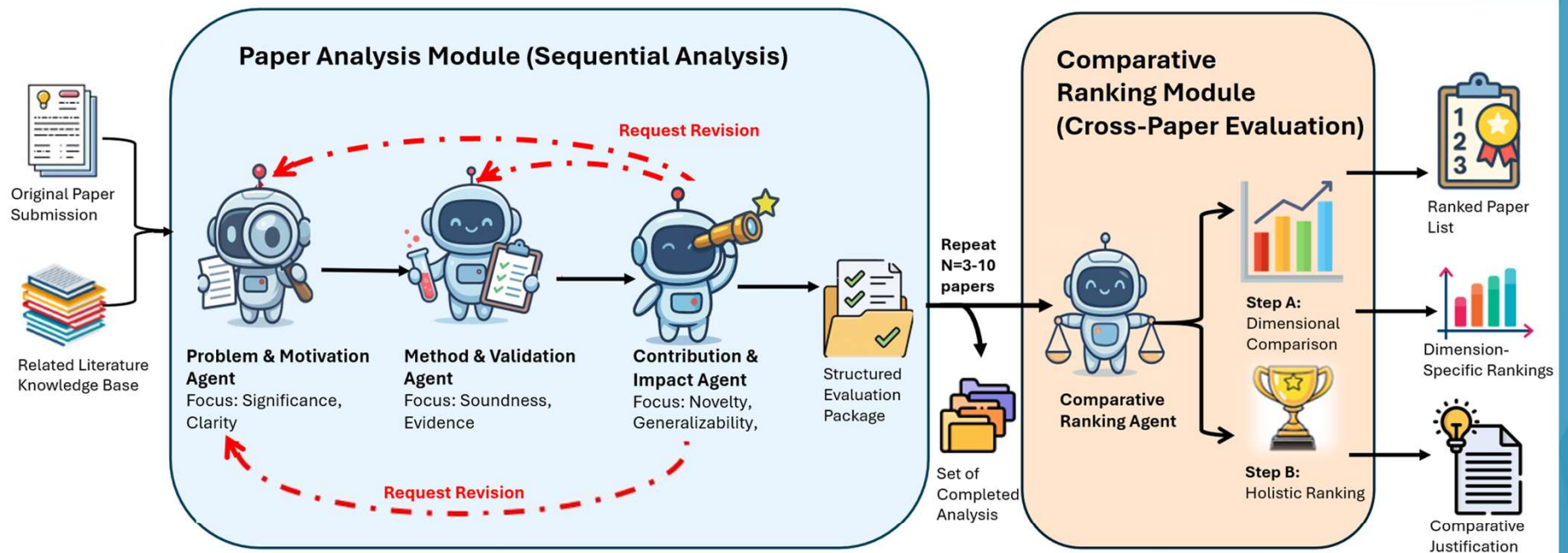


Figure: Modular Pipeline Architecture of ProComp Incorporating Structured Preprocessing, Specialized Evaluation Agents, Meta-Review Fusion, and Retrieval-Augmented Reasoning

Agents Architecture



Problem Analysis Agent: Evaluates clarity, relevance, and motivation of the research problem. Focuses on how well the paper frames its scientific questions and research goals, providing numerical scores and explanations.



Methodology Agent : Assesses technical soundness, experimental rigor, and reproducibility of methods used. Ensures the methodology aligns with research objectives and is validated with appropriate experiments



Contribution Agent : Measures novelty, significance, and potential impact of the research. Distinguishes high-impact work by evaluating the originality and broader relevance of the contributions



Meta-Review Agent: Aggregates scores and qualitative feedback from other agents to produce a final, coherent evaluation. Resolves conflicts and provides overall ranking and recommendations.

RAG

Integrated RAG with Semantic Scholar Retrieval

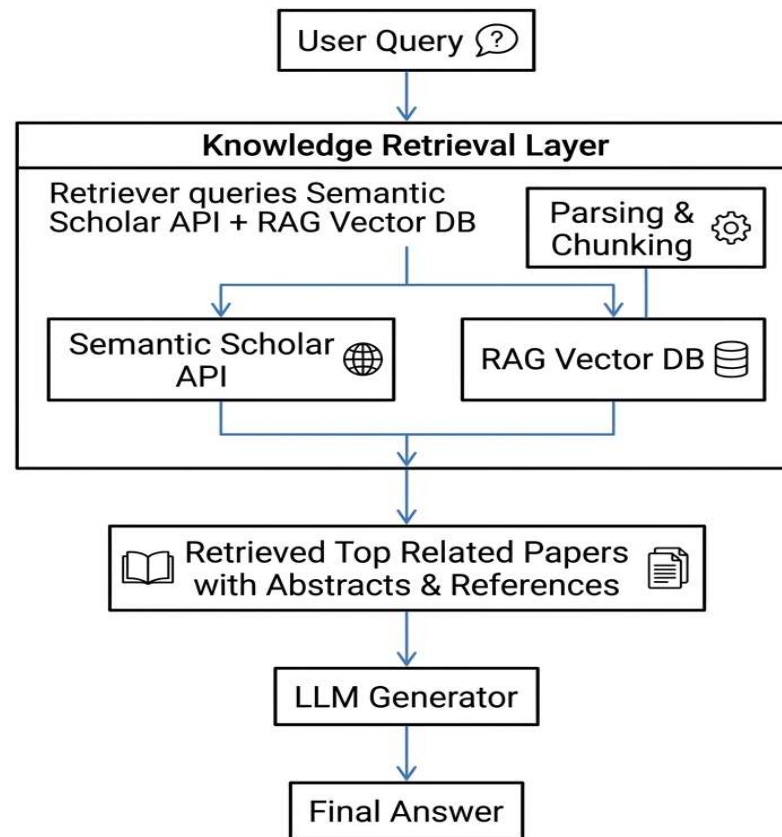


Figure :Integrated Retrieval-Augmented Generation Pipeline with Semantic Scholar API for Academic Query Resolution

Prompt engineering Strategy



Role-Specific Structured Prompts



Chain-of-Thought Reasoning



Integration of Retrieval Context

Experimental setup

Curated Peer-Review Corpus

The dataset comprises 194 scientific manuscripts modeled after conference-style peer review, with structured summaries and detailed human-evaluation scores across key dimensions such as soundness, presentation, and contribution.

Heterogeneous Research Domains

Papers cover a broad spectrum of fields including AWA1, AWA2, CUB, SUN, and aPY datasets, ensuring the system's robustness and generalizability across diverse scientific topics and domain shifts.

Multi-Dimensional Annotations

Each paper includes normalized human scores for various criteria, enabling fine-grained analysis of quality. Annotations cover methodological rigor, clarity, novelty, and overall impact, reflecting the complexity of academic evaluation.

Model Configurations



LLM Models Tested

The framework was tested on GPT-based, LLaMA-based, NVIDIA Nemotron, and Qwen models accessed via OpenRouter API to compare capabilities across architectures.



Inference Settings

Controlled inference with low temperature, fixed token limits, standardized sampling, and consistent request frequency ensured reproducibility and minimized output variability.



Execution Environment

Pipeline orchestration and evaluation metrics ran locally while model inference leveraged cloud-based OpenRouter API, enabling efficient large-scale testing.

Evaluation metrics

▶ Error-Based Metrics

- ❑ Mean Absolute Error (MAE)
- ❑ Root Mean Squared Error (RMSE)

▶ Correlation-Based Metrics

- ❑ Pearson Correlation
- ❑ Spearman Correlation

▶ Ranking-Based Evaluation

- ❑ Pairwise Ranking
- ❑ Kendall's Tau

▶ Variance for Within-Group

- ❑ Standard Variance



Comparative analysis of Multi-Agent framework

- ▶ Baselines Frameworks
- ▶ Holistic Ranking Evaluation
- ▶ Correlation analysis
- ▶ Ablation study



Baselines Framework

Rule-Based Semantic Evaluation

SEA-E: Uses heuristic scoring and semantic similarity to evaluate research papers based on predefined rules. It offers consistent outputs but lacks a nuanced understanding of novelty and methodology.

Structured Pre-Agentive Review Frameworks

TreeReview : Organizes paper review as a hierarchical decision tree with clear modularity and interpretability but is limited by rigid rules and lacks advanced reasoning capabilities.

Baselines Framework

Early Single-Agent LLM Reviewer

Zero-Shot Prompting : It enables a model to perform evaluation tasks without relying on task-specific training examples, instead leveraging its pre-trained knowledge and generalization capability. In the context of peer review, it generates direct assessments based solely on the input manuscript but may lack depth and consistency due to the absence of guided reasoning or structured decomposition.

Chain-of-Thought (CoT) Prompting: It encourages the model to produce explicit intermediate reasoning steps before arriving at a final decision, thereby improving performance on complex analytical tasks. In evaluation settings, it enhances interpretability and reasoning quality; however, it can still produce inconsistent or verbose outputs without structured control mechanisms.

Baselines Framework

Agent Review

Agent Review uses structured prompting to simulate multiple evaluation dimensions within a single framework, enabling a more organized, criterion-based assessment than basic prompting methods.

Deep Reviewer

The reviewer focuses on generating detailed, in-depth analytical feedback by encouraging comprehensive reasoning, aiming to improve evaluation quality through richer, more elaborate review outputs.

Cycle Reviewer

Uses multiple evaluation iterations to refine outputs, improving consistency, but is limited by scalability and unable to incorporate independent agent-based cross-validation.

Interpretation of Baselines evaluated with Error-Based Metrics

System	Type	MAE	RMSE
Baseline	sea_e	1.1490	1.4778
	cycle_reviewer	1.2132	1.5671
	treereview	1.2211	1.5684
	agentreview	1.2600	1.6179
	deep_reviewer	1.2735	1.6371
	zero_shot	1.3184	1.6982
	chain_of_thought	1.4543	1.8430
MainModel	GPT	0.9744	1.2239
	NVIDIA	1.2971	1.5394

Table : Performance metrics (MAE and RMSE) for all evaluated systems on the 194-sample review dataset.

HOLISTIC RANKING EVALUATION

Score Variance Analysis Across Group Type

Setting	Model	Variance
Homogeneous	GPT	0.2640
	QWEN	0.4056
	Human	0.0147
Heterogeneous	GPT	2.0199
	QWEN	0.7562
	Human	1.4266

Table: Score Variance Across Homogeneous and Heterogeneous Groups

Ranking Correlation with Human Judgments (Kendall's Tau)

Setting	Model	Kendall's Tau
Homogeneous	GPT	0.76
	LLAMA	0.52
	QWEN	0.66
Heterogeneous	GPT	0.58
	LLAMA	0.82
	QWEN	0.78

Table : Kendall's Tau Correlation with Human Rankings

HOLISTIC RANKING EVALUATION

Pairwise Ranking Accuracy

Setting	Model	Pairwise Accuracy vs Human Scores
Homogeneous	GPT	0.63
	LLAMA	0.51
	QWEN	0.58
Heterogeneous	GPT	0.54
	LLAMA	0.66
	QWEN	0.64

Table: Pairwise Ranking Accuracy Across Homogeneous and Heterogeneous Groups

Within-Group Ranking Analysis

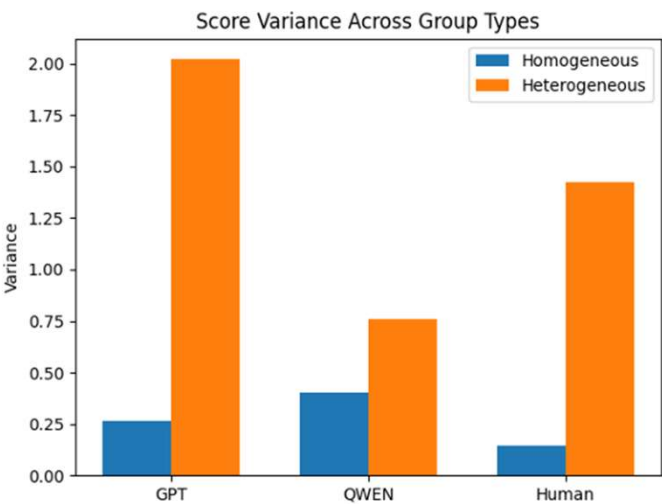


Figure: Score variance across homogeneous and heterogeneous group types.

Within-Group Ranking Analysis

Group	Paper ID	Overall	Motivation	Method	Contribution	Rank
Homogeneous	KHX0dKXdqH	7.78	8.67	7	7.67	1
	IUKff7nYmW	7.89	8.67	7	8.00	2
	a1wf2N967T	7.67	8.00	7	8.00	3

Justification: IUKff7nYmW ranks first because it pairs the strongest motivation with the highest contribution, giving it the clearest overall impact despite the tight score spread. KHX0dKXdqH follows closely with equally strong motivation but slightly lower contribution. a1wf2N967T is solid but shows the weakest motivation of the three.

Heterogeneous	gojL67CfS8	9.00	8.00	7	8.00	1
	A0HSmrwtLH	6.00	5.00	5	8.00	2
	MXzr10iX2d	5.67	7.00	5	5.00	3

Justification: gojL67CfS8 clearly explains the problem, motivates why it matters, and offers a strong contribution. A0HSmrwtLH has moderate motivation but a solid contribution. MXzr10iX2d lacks strong motivation and provides limited contribution.

Comparative Group-Wise Rankings with Dimensional Scores and Justifications

CORRELATION ANALYSIS EVALUATION

System	Pearson r	Spearman ρ
GPT	0.862	0.634
LLAMA	0.786	0.600
NVIDIA	0.781	0.585
QWEN	0.898	0.665
AgentReview	0.398	0.400
Chain_of_Thought	0.353	0.329
Cycle_Reviewer	0.362	0.354
Deep_Reviewer	0.399	0.379
SEA_E	0.392	0.368
TreeReview	0.359	0.318
Zero_Shot	0.416	0.425

Table: Correlation of Model Predictions with Human Scores Across Main Models and Baselines

ABLATION STUDY

The following are the ablation experiments performed to understand each agent's influence on our multi-agent framework results.

- ▶ With Problem Analysis Agent only
- ▶ With Methodology Agent only
- ▶ With Contribution Agent only
- ▶ Without Problem Analysis Agent
- ▶ Without Methodology Agent
- ▶ Without Contribution Agent
- ▶ Without Meta-Review Agent
- ▶ Impact of Retrieval Augment Generation (RAG)

Ablation Study Result

Ablation Name	MAE	RMSE
Without Retrieval (RAG)		
With Problem Analysis Agent only	0.933	1.134
With Methodology Agent only	1.357	1.615
With Contribution Agent only	1.280	1.523
Without Problem Analysis Agent	1.476	1.695
Without Methodology Agent	1.351	1.621
Without Contribution Agent	1.357	1.642
Without Meta-Review Agent	1.141	1.303
With Retrieval (RAG)		
With Problem Analysis Agent only	0.953	1.184
With Methodology Agent only	1.258	1.486
With Contribution Agent only	1.222	1.465
Without Problem Analysis Agent	1.466	1.669
Without Methodology Agent	1.148	1.411
Without Contribution Agent	1.153	1.430
Without Meta-Review Agent	1.041	1.224
Full Multi-Agent System	0.653	0.846

Ablation Study of Multi-Agent Components With and Without Retrieval (RAG)
Using MAE and RMSE Metrics

Conclusion

- ▶ The proposed ProComp framework addresses the most important challenges present in existing and existing peer-review based practices and introduces it through a multi-agent evaluation paradigms through a structured and multi-agent evaluation framework that can be used effectively to decompose the review process into distinct components that make different research aspects in a way that would allow for the more granularity, interpretability and alignment through multiple research aspects of the entire reviews process.
- ▶ ProComp is supported by Retrieval-Augmented Generation (RAG), which supports strong factual grounding, reduces hallucinations, and helps to reinforce automated evaluations. JSON-based outputs are more usable for accurately assessing an automated review, with the benefit of being easily peer-reviewed and easily compared with a human.
- ▶ ProComp outperforms existing baselines in accuracy, reliability, data quality, and ranking ability, and also demonstrates a significant difference in research quality. Our framework works well alongside human judgment, which supports its use.
- ▶ ProComp offers a scalable, robust peer-review solution with high in-house automation and interpretability, helping an intelligent review team support human reviewers in making fair, efficient, and high-quality decisions throughout the process.

Limitations and Challenges

- ▶ ProComp is dependent on underlying LLM capabilities, making it vulnerable to issues such as hallucinations, prompt sensitivity, and variability across different models.
- ▶ The effectiveness of Retrieval-Augmented Generation (RAG) is contingent on the quality and relevance of retrieved sources, which may introduce noise or bias into the evaluation process.
- ▶ The multi-agent architecture increases computational overhead and system complexity, resulting in higher latency and resource consumption for large-scale deployment.
- ▶ Alignment with human judgment is not perfect, and maintaining consistency across diverse and highly specialized research domains remains a challenging task

Future work

- ▶ We plan to improve the robustness and generalization of ProComp by leveraging advanced LLMs and employing prompt design strategies to ensure consistent, accurate evaluation.
- ▶ Improvements to the Retrieval-Augmented Generation (RAG) component, with domain-specific and dynamically updated knowledge sources, could provide more solid grounding in the field and reduce incorrect evaluations.
- ▶ We will also explore advanced judge mechanisms and competitive discussion protocols among agents, incorporating agreement-based scoring, consensus modeling, and pairwise ranking strategies to improve reliability, calibration, and discriminative ranking performance.
- ▶ Trying to optimize the multi-agent architecture to reduce computational overhead, latency, and resource consumption for large-scale implementation.
- ▶ Extending the framework to various research domains and multilingual evaluation will broaden the scope and the size of that methodology in real-world peer-review scenarios.

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Thank you